

Coding of geodesic flow on geometrically finite hyperbolic 3-manifolds with a cusp

1 Coding geodesic flow and limit set

Setup: Let Γ be a geometrically finite subgroup of $\mathrm{SL}_2(\mathbb{R})$ or $\mathrm{SL}_2(\mathbb{C})$ such that $M := \Gamma \backslash \mathbb{H}^d$ (here $d = 2$ or 3) with only one cusp. Suppose $\infty \in \partial \mathbb{H}^d$ is a parabolic fixed point, and let $\Gamma_\infty = \mathrm{Stab}_\infty(\Gamma)$ be the associated parabolic subgroup. We choose Δ_0 a fundamental domain of $\partial \mathbb{H}^d - \{\infty\} \simeq \mathbb{R}^{d-1}$ of Γ_∞ . Let Λ be the limit set of Γ on $\partial \mathbb{H}^d$ and $\Lambda_0 = \Lambda \cap \Delta_0$. Let μ be a Paterson-Sullivan measure supported on Δ . Let δ be the critical exponent.

Recall that we have the Hopf parameterization of the geodesic flow given by $(\partial \mathbb{H}^d \times \partial \mathbb{H}^d - \mathrm{Diag}) \times \mathbb{R} \simeq T^1 \mathbb{H}^d$. With a correct choice of identification, we have the action of Γ is given by

$$\gamma(x, y, t) = (\gamma x, \gamma y, t + \log |D\gamma(x)|), \quad (1.1)$$

where $\log |D\gamma(x)|$ is cohomologous to the usual Busemann function.

We first explain how to obtain a coding of geodesic flow from Proposition 1.1. Here is the main coding result from [LP23] is the following

Proposition 1.1. *There exists a family of disjoint open sets $\{\Delta_j\}_{j \geq 1}$ and an expanding map T on $\cup \Delta_j$ such that*

- $\mu(\Delta_0) = \sum \mu(\Delta_j)$
- For each j , there exists $\gamma_j \in \Gamma$ such that $T|_{\Delta_j} = \gamma_j^{-1}$ and $\gamma_j \Delta_0 = \Delta_j$
- There exists $\lambda < 1$ such that for all j , $|D\gamma_j| \leq \lambda$. (where $|D\gamma_j|$ is the norm of the derivative to the euclidean metric on $\mathbb{R}^{d-1}$)
- (Exponential tail) There exists $\epsilon > 0$ such that

$$\sum_j |D\gamma_j|^{\delta - \epsilon} < \infty.$$

Indeed

$$R(x) = \log |D\gamma_j^{-1}(x)| \quad (1.2)$$

for $x \in \Delta_j$ is the return time and the exponential tail can also be expressed as

$$\int_{\Delta_0} e^{\epsilon R(x)} d\mu(x) < \infty$$

Before going to the coding of flow, we need to add a stable direction to obtain a section of the flow. This is another proposition, which is not hard

Proposition 1.2. *There exists $\Lambda_- \subset \Lambda$ near ∞ such that for all γ_j , we have $\gamma_j^{-1} \Lambda_- \subset \Lambda_-$.*

Fix this identification, we let

$$S = (\cup \Delta_j) \times \Lambda_- \times \{0\}$$

be a section on $T^1\mathbb{H}^d$, which also descends to a section in T^1M . Denote the map by π (Add a graph) **Attention: The coding system we construct later is possibly larger than the geodesic flow. We are not using the first return map to the section.**

We can construct a suspension flow on S by

$$S^R := S \times \mathbb{R}^+ / (x, y, t + R(x)) \sim (Tx, T(x, y), t) \quad (1.3)$$

where for $x \in \Delta_j$, we have $Tx = \gamma_j^{-1}x \in \Delta_0$ and $T(x, y) = \gamma_j^{-1}y \in \Lambda_-$ due to Proposition 1.2. On S^R we have a semiflow on the $\mathbb{R}^+$ direction, denoted by T_t .

Here is the connection between the suspension flow and the geodesic flow (where ν is T -invariant and absolute continuous to the Patterson-Sullivan, comes from general theory of expanding systems or Gibbs measure)

Lemma 1.3. *The map π induces a factor map (S^R, T_t, ν^R) to $(T^1M, \mathcal{G}_t, m^{BMS})$, that is*

$$\pi \circ T_t = \mathcal{G}_t \circ \pi, \quad \pi_*(\nu^R) = m^{BMS}$$

Soft part: The equations Eq. (1.3), Eq. (1.1) and Eq. (1.2) imply that π well defined, since the equivalence relation is mapped to equivalence relation under Γ action, we have

$$\gamma_j(Tx, T(x, y), t) = \gamma_j(\gamma_j^{-1}x, \gamma_j^{-1}y, t) = (x, y, t + \log |D\gamma_j^{-1}(x)|) = (x, y, t + R(x))$$

Since the geodesic flow is mixing, almost every geodesic trajectory will cross this transversal section $\pi(S)$ of T^1M .

Hard part: The proof of equality of measures uses that the T -invariant measure ν on Λ_0 is absolutely continuous to the Patterson-Sullivan measure μ and the product structure, please see [LP23, Prop. 4.15]

Remark 1.4. The delicate thing here is that the map π from S^R to T^1M is not injective, since R may not be the first return time to the section. We only obtain a factor map, but it is still sufficient to conclude the exponential mixing of geodesic flow from suspension flow, see [LP23, Section 4.2]. For the suspension flow, we use a modified Dolgopyat's argument to prove its exponential mixing.

Since we don't need the first return to the section, the advantage is that the construction of coding is more flexible. See also Young tower for a similar situation in the map case.

2 Limit set coding for $SL_2(\mathbb{R})$ and $SL_2(\mathbb{C})$

We start to construct coding of Proposition 1.1. We work only on the boundary $\partial\mathbb{H}^d$. The idea is to use "flowers" near parabolic fixed points. The advantage is that the boundary of flower is clean, which enables us to get good measure estimate to obtain exponential tail.

Let H_0 be the horoball based at infinity. Let $p = \gamma\infty$ be a parabolic fixed point in Δ_0 and $H_p = \gamma H_0$ a horoball based at p . Suppose $\gamma^{-1}\infty \in \Delta_0$. Let h_p be the height of H_p , that is the euclidean height in the upper half space model.

We can tile a neighborhood of p using a flower J_p : Let $\eta > 0$ be some small parameter. Let F_∞ be the complement of a large parallelotope of size $1/\eta$ tiled by Δ_0 using Γ_∞ . Let

$$N_p = \{\gamma h \in \Gamma_\infty : h \in \Gamma_\infty, h\Delta_0 \notin F_\infty\}, \quad J_p = \gamma F_\infty^c = \bigcup \{\gamma h \Delta_0 : \gamma h \in N_p\}.$$

By computation, we can have

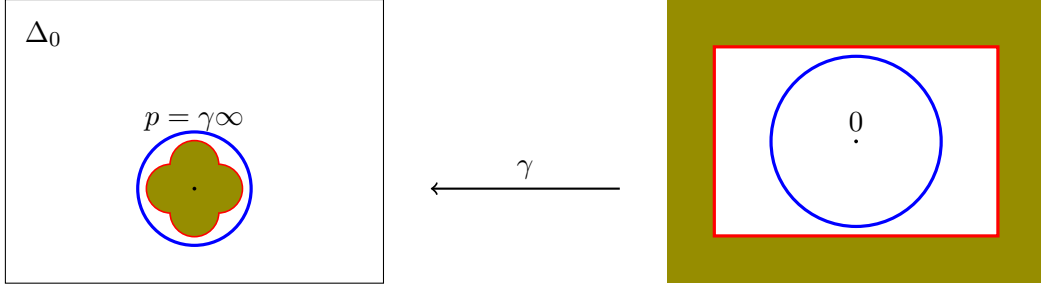


Figure 1: The shaded region on the left hand side is J_p , which is the image under the action of γ on the complement of the white rectangle on the right hand side.

Lemma 2.1. *There exists $C > 0$ such that for any parabolic fixed point $p \in \Delta_0$, we have*

$$B(p, \eta h_p / C) \subset J_p \subset B(p, C \eta h_p)$$

Remark 2.2. For non-maximal rank case, we need modify a bit the definition of J_p . Since the direction without translation action of Γ_∞ is bounded, we take a large set to cover it.

By this way, we find a J_p near a parabolic fixed point and it is a building block of coding. In order to code almost all the set Λ_0 , we need to use flowers of decreasing size. But we cannot be too greedy, we will only take flowers a bit away from the boundary to make sure the remaining set still has good measure estimate.

Here is the way:

- Let $\Omega_0 = \Delta_0$
- For $n \in \mathbb{N}$, let P_n be the set of parabolic fixed points in Ω_n such that $-\log(\eta h_p) \in [n, n+1]$ and $d(p, \partial\Omega_n) > h_p$. (J_p has size e^{-n} , its distance to the boundary is of size e^{-n}/η)
- Let

$$\Omega_{n+1} = \Omega_n - \bigcup_{p \in P_n} J_p$$

The key proposition is

Proposition 2.3. *There exists $C' > 0$, $\epsilon > 0$ such that*

$$\mu(\Omega_n) \leq C'(1 - \epsilon)^n$$

From this proposition, we can take

$$\{\Delta_j\}_{j \geq 1} = \{\gamma \Delta_0 : \gamma \in N_p, p \in P_n, n \in \mathbb{N}\}$$

3 Energy exchange and exponential tail

This part is devoted to prove Proposition 2.3. We need to go back to the flow and manifold point of view to obtain more information of these flowers. The key ingredients are: Separation of parabolic fixed points, recurrence of geodesic flow and energy exchange argument (interesting point). (non-concentration and away from boundary, for the expert)

Due to geometrically finite, we know H_p are disjoint for different p .

References

- [LP23] Jialun Li and Wenyu Pan. Exponential mixing of geodesic flows for geometrically finite hyperbolic manifolds with cusps. *Inventiones mathematicae*, 231(3):931–1021, March 2023.